

GENOME REARRANGEMENTS AND SORTING BY REVERSALS*

VINEET BAFNA[†] AND PAVEL A. PEVZNER[‡]

Abstract. Sequence comparison in molecular biology is in the beginning of a major paradigm shift—a shift from *gene* comparison based on local mutations (i.e., insertions, deletions, and substitutions of nucleotides) to *chromosome* comparison based on global rearrangements (i.e., inversions and transpositions of fragments). The classical methods of sequence comparison do not work for global rearrangements, and little is known in computer science about the edit distance between sequences if global rearrangements are allowed. In the simplest form, the problem of gene rearrangements corresponds to *sorting by reversals*, i.e., sorting of an array using reversals of arbitrary fragments. Recently, Kececioğlu and Sankoff gave the first approximation algorithm for sorting by reversals with guaranteed error bound 2 and identified open problems related to chromosome rearrangements. One of these problems is Gollan’s conjecture on the reversal diameter of the symmetric group. This paper proves the conjecture. Further, the problem of expected reversal distance between two random permutations is investigated. The reversal distance between two random permutations is shown to be very close to the reversal diameter, thereby indicating that reversal distance provides a good separation between related and unrelated sequences in molecular evolution studies. The gene rearrangement problem forces us to consider reversals of *signed permutations*, as the genes in DNA could be positively or negatively oriented. An approximation algorithm for signed permutation is presented, which provides a performance guarantee of $\frac{3}{2}$. Finally, using the signed permutations approach, an approximation algorithm for sorting by reversals is described which achieves a performance guarantee of $\frac{7}{4}$.

Key words. computational molecular biology, sorting by reversals, genome rearrangements

AMS subject classifications. 68Q25, 68Q05

1. Introduction. Genus *Lobelia* comprises over 350 species that range from small, slender herbs to woody, giant-rosette plants. Figure 1 presents the order of genes in *Tobacco* and *Lobelia fervens* chloroplast genomes with a hypothetical sequence of rearrangement events (Knox et al. [KDP93]) during evolution of *Lobelia fervens* from a tobacco-like ancestral genome.

It is not so easy to verify that the evolutionary events shown in Fig. 1 represent the *shortest* series of reversals transforming the *Tobacco* permutation into the *Lobelia fervens* permutation. In fact, Theorem 2 of this paper indicates that the shortest sequence of rearrangement events contains just 4 reversals, shown in Fig. 2 (however, for the case of *signed* permutations (see below) the evolutionary events presented in Fig. 1 do represent the shortest series of reversals).

With the advent of large-scale DNA mapping and sequencing, the number of *genome comparison* problems similar to the one presented in Fig. 1 is rapidly growing in different areas, including evolution of plant cpDNA (Raubeson and Jansen [RJ92], Hoot and Palmer [HP94]) and mtDNA (Palmer and Herbon [PH88], Faure and Havlik [FH89]), animal mtDNA (Hoffman et al. [HBB92], Sankoff et al. [SLA92]), virology (Koonin and Dolya [K93], Hannenhalli et al. [HCKP95]), *Drosophila* genetics (Whiting et al. [WPFJ89]), and comparative physical mapping (Lyon [L88]). Genome comparison has certain merits and demerits as compared to classical *gene* comparison.

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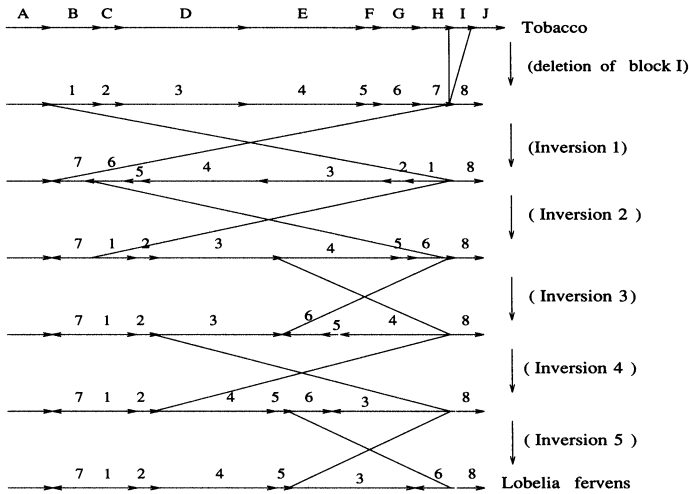


FIG. 1. Evolution of *Lobelia fervens*.

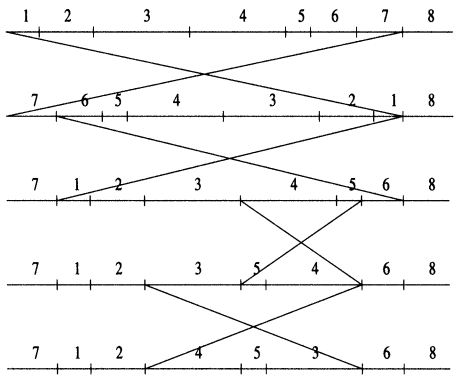


FIG. 2. An optimal reversal sequence for $\pi = 71245368$.

Genome comparison ignores actual DNA sequences of genes, while gene comparison ignores gene order. The ultimate goal would be to combine merits of both genome and gene comparison in a single algorithm. However, until recently, studies of genome rearrangements were based on heuristic methods, and there were no algorithms to analyze gene orders in molecular evolution. Kececioglu and Sankoff found an algorithm for reversal distance with guaranteed error bound 2 and raised a spectrum of open problems motivated by genome rearrangements [KS93]. The present paper solves some of them.

In the problem we consider, the order of genes in two organisms is represented by permutations $\pi = (\pi_1\pi_2\ldots\pi_n)$ and $\sigma = (\sigma_1\sigma_2\ldots\sigma_n)$. A reversal $\rho(i, j)$ of an interval $[i, j]$ is the permutation

$$\begin{pmatrix} 1 & 2 & \dots & i-1 & i & i+1 & \dots & j-1 & j & j+1 & \dots & n \\ 1 & 2 & \dots & i-1 & j & j-1 & \dots & i+1 & i & j+1 & \dots & n \end{pmatrix}.$$

Clearly $\pi \cdot \rho$ has the effect of reversing genes $\pi_i, \pi_{i+1}, \dots, \pi_j$.

Given permutations π and σ , the *reversal distance problem* is to find a series of reversals $\rho_1, \rho_2, \dots, \rho_t$ such that $\pi \cdot \rho_1 \cdot \rho_2 \cdots \rho_t = \sigma$ and t is minimum. We call t the *reversal distance* between π and σ . Note that reversal distance between π and σ equals the reversal distance between $\sigma^{-1}\pi$ and the *identity* permutation ι . *Sorting π by reversals* is the problem of finding reversal distance $d(\pi)$, between π and ι .

Reversals generate the *symmetric group* S_n . Given an arbitrary permutation π from S_n , we seek a shortest product of *generators* $\rho_1 \cdot \rho_2 \cdots \rho_t$ that equals π . Even and Goldreich [EG81] show that given a set of generators of a permutation group and a permutation π , determining the shortest product of generators that equals π is NP-hard. Jerrum [J85] proves that the problem is PSPACE-complete, and remains so, when restricted to two generators. In our problem, the generator set is fixed. However, Kececioglu and Sankoff [KS93] conjecture that sorting by reversals is NP-complete.

Gates and Papadimitriou [GP79] studied a similar *sorting by prefix reversals* problem (also known as *pancake-flipping problem*): given an arbitrary permutation π , find $d_{pref}(\pi)$, which is the minimum number of reversals of the form $\rho(1, i)$ that sort π . Their concern is with bounds on the *prefix reversal diameter* of the symmetric group, $d_{pref}(n) = \max_{\pi \in S_n} d_{pref}(\pi)$. They show that $d_{pref}(n) \leq \frac{5}{3}n + \frac{5}{3}$ (see also [GT78]) and that for infinitely many n , $d_{pref}(n) \geq \frac{17}{16}n$ [GP79]. Aigner and West [AW87] consider the diameter of sorting when the operation is reinsertion of the first element, and Amato et al. [ABSR89] consider a variation inspired by reversing trains. Kececioglu and Sankoff [KS93] found an approximation algorithm for sorting by reversals with performance guarantee 2. They also devised efficient bounds, allowing them to solve the reversal distance problem optimally or almost optimally for n ranging from 30 to 50. This range covers the biologically important case of mitochondrial genomes.

Define $d(n) = \max_{\pi \in S_n} d(\pi)$ to be the *reversal diameter* of the symmetric group of order n . Gollan conjectured that $d(n) = n - 1$ and that only one permutation γ_n , and its inverse, γ_n^{-1} , require $n - 1$ reversals to be sorted (see Kececioglu and Sankoff [KS93] for details). The *Gollan* permutation, in one-line notation, is defined as follows:

$$\gamma_n = \begin{cases} (3, 1, 5, 2, 7, 4, \dots, n-3, n-5, n-1, n-4, n, n-2), & n \text{ even,} \\ (3, 1, 5, 2, 7, 4, \dots, n-6, n-2, n-5, n, n-3, n-1), & n \text{ odd.} \end{cases}$$

For $n \leq 11$, Gollan verified this conjecture using extensive computations. Kececioglu and Sankoff [KS93] developed lower bounds for reversal distance, allowing them to verify Gollan's conjecture for $n \leq 200$ for $n \equiv 1 \pmod{3}$. In the present paper, we introduce the notion of the *breakpoint graph* of a permutation and establish the links between reversal distance and *maximum cycle decomposition* of this graph. This construction allows us to prove Gollan's conjecture. Further, we study the problem of expected reversal distance between two random permutations. We demonstrate that reversal distance between two random permutations is very close to the reversal diameter of the symmetric group, thereby indicating that reversal distance provides a good separation between related and nonrelated sequences in molecular evolution studies.

Afterwards, we study reversals of *signed* permutations. The *Lobelia fervens* permutation (Fig. 1) corresponds to the signed permutation $(-7, +1, +2, +4, +5, +3, -6, +8)$. In the biologically more relevant signed case, every reversal of fragment $[i, j]$ changes the signs of the elements within that fragment. We are interested in the minimum number of reversals required to transform the signed permutation π into the identity signed permutation $(+1, +2, \dots, +n)$. We devise an approximation algorithm



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to v equals the number of gray edges incident to v . A *balanced graph* is a graph in which every vertex is balanced. It is easy to see that $G(\pi)$ is a balanced graph for every π ; therefore, it contains an alternating Eulerian cycle in every connected component. For characterization of alternating Eulerian cycles in edge-colored graphs, see Kotzig [K68] and Pevzner [P94].

Cycle decompositions play an important role in estimating the reversal distance. When we apply a reversal to a permutation, there might be a change in the number of breakpoints, as well as in the number of cycles in a maximum decomposition. In Theorem 1, we show that there is a strong correlation between these two changes. This idea allows us to bound the reversal distance in terms of the size of the maximum cycle decomposition.

Denote the number of black edges in $G(\pi)$ (breakpoints in π) as $b = b(\pi)$ and the number of adjacencies in π as $a(\pi)$. Let $c(\pi)$ be the number of cycles in a maximum cycle decomposition of $G(\pi)$. Given an arbitrary reversal ρ , denote $\Delta b = \Delta b(\pi, \rho) = b(\pi\rho) - b(\pi)$ (increase in breakpoints), $\Delta a = \Delta a(\pi, \rho) = a(\pi\rho) - a(\pi)$ (increase in adjacencies), and $\Delta c = \Delta c(\pi, \rho) = c(\pi\rho) - c(\pi)$ (increase in the number of cycles in a maximum decomposition).

THEOREM 1. *For every permutation π and reversal ρ , $\Delta c(\pi, \rho) - \Delta b(\pi, \rho) \leq 1$.*

Proof. We augment $G(\pi)$ to get $G'(\pi)$ as follows: For every adjacency (π_i, π_{i+1}) in π , add a cycle of length 2 with one black and one gray edge connecting π_i and π_{i+1} . Every decomposition of $G(\pi)$ into $c(\pi)$ cycles corresponds to a decomposition of $G'(\pi)$ into $c(\pi) + a(\pi)$ cycles and, conversely, any decomposition of $G'(\pi)$ into $c'(\pi)$ cycles corresponds to a decomposition of $G(\pi)$ into at least $c'(\pi) - a(\pi)$ cycles.

We will prove that the number of cycles in $G'(\pi)$ changes by no more than one in a single reversal. An arbitrary reversal $\rho(i, j)$ involves 4 vertices of $G'(\pi)$ and leads to replacing two black edges $DEL = \{(\pi_{i-1}, \pi_i), (\pi_j, \pi_{j+1})\}$ by the black edges $ADD = \{(\pi_{i-1}, \pi_j), (\pi_i, \pi_{j+1})\}$.

If these two black edges in ADD belong to the same cycle in a maximum cycle decomposition of $G'(\pi\rho)$, then a deletion of that cycle gives a cycle decomposition of $G'(\pi)$ with at least $c'(\pi\rho) - 1$ cycles. Therefore, $c'(\pi) \geq c'(\pi\rho) - 1$ and $\Delta c' = \Delta c + \Delta a \leq 1$.

On the other hand, if the black edges in ADD belong to different cycles C_1 and C_2 in a maximum cycle decomposition of $G'(\pi\rho)$, then deleting $C_1 \cup C_2$ gives a set of edge-disjoint cycles of size $c'(\pi\rho) - 2$ in the graph $G(\pi\rho) \setminus (C_1 \cup C_2)$. Clearly, the set of edges $(C_1 \cup C_2 \cup DEL) \setminus ADD$ forms a balanced graph and must contain at least one cycle. Combining this cycle with the previously obtained $c'(\pi\rho) - 2$ cycles, we obtain a cycle decomposition of $G(\pi) = (G(\pi\rho) \setminus (C_1 \cup C_2)) \cup (C_1 \cup C_2 \cup DEL \setminus ADD)$ into at least $c'(\pi\rho) - 1$ cycles. Therefore, $\Delta c' = \Delta c + \Delta a \leq 1$.

Finally, observing that $a(\pi) + b(\pi) = n + 1$ implies $\Delta b + \Delta a = 0$, we prove the theorem. $\square$

Theorem 1 immediately gives us a new lower bound for the reversal distance in terms of the number of breakpoints and the size of a maximum cycle decomposition of the breakpoint graph.

THEOREM 2. *For every permutation π , $d(\pi) \geq b(\pi) - c(\pi)$.*

Proof. Let $\rho_t, \dots, \rho_1$ be a shortest series of reversals transforming $\pi = \pi_t$ into the identity permutation π_0 . Denote $\pi_{i-1} = \pi_i \rho_i$ for $i = 1, \dots, t$, and apply Theorem 1 for a permutation π_i and reversal ρ_i .

$$d(\pi_i) = d(\pi_{i-1}) + 1 \geq d(\pi_{i-1}) - \Delta b(\pi_i, \rho_i) + \Delta c(\pi_i, \rho_i)$$

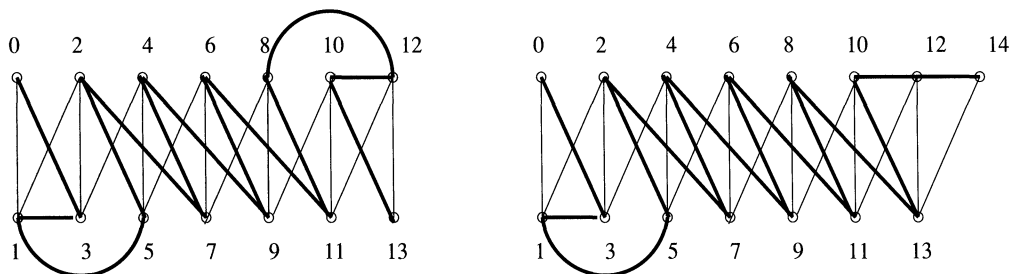


FIG. 5. $G(\gamma_{12})$ and $G(\gamma_{13})$.

$$= d(\pi_{i-1}) + (b(\pi_i) - b(\pi_{i-1})) + (c(\pi_{i-1}) - c(\pi_i)).$$

Recalling that $d(\pi_0) = b(\pi_0) = c(\pi_0) = 0$, we get

$$d(\pi_i) - (b(\pi_i) - c(\pi_i)) \geq d(\pi_{i-1}) - (b(\pi_{i-1}) - c(\pi_{i-1})) \geq \cdots \geq d(\pi_0) - (b(\pi_0) - c(\pi_0)) = 0.$$

Substituting $i = t$, we prove the theorem. $\square$

3. Reversal diameter of the symmetric group. Now we have a characterization of the reversal distance of a permutation in terms of the maximum cycle decomposition of the breakpoint graph. Next we show that the graph corresponding to the Gollan permutation, γ_n , has at most two disjoint alternating cycles.

LEMMA 1. *Every alternating cycle in $G(\gamma_n)$ contains the vertex 1 or 3.*

Proof. Let us recall the Gollan permutation:

$$\gamma_n = \begin{cases} (3, 1, 5, 2, 7, 4, \dots, n-3, n-5, n-1, n-4, n, n-2), & n \text{ even,} \\ (3, 1, 5, 2, 7, 4, \dots, n-6, n-2, n-5, n, n-3, n-1), & n \text{ odd.} \end{cases}$$

Let i be the minimal odd vertex of an alternating cycle X in $G(\gamma_n)$. Consider the sequence i, j, k of consecutive vertices in X , where (i, j) is black, and (j, k) is gray.

If $i > 5$, then $j = i - 3$ or $j = i - 5$ and $k = j + 1$ or $k = j - 1$ (see the structure of the Gollan permutation and Fig. 5), implying that k is odd and $k < i$, a contradiction. If $i = 5$, then $j = 2$ and k is either 1 or 3, a contradiction. Therefore, i is either 1 or 3. $\square$

THEOREM 3 (Gollan conjecture). *For every n , $d(\gamma_n) = d(\gamma_n^{-1}) = n - 1$.*

Proof. For $n \leq 2$, the claim is trivial. For $n > 2$, partition the vertex set of $G(\gamma_n)$ into $V_l = \{0, 1, 3\}$ and V_r . From Lemma 1 and the fact that there is no cycle contained in V_l , we see that every alternating cycle must contain at least 2 edges from the cut (V_l, V_r) . Because the cut (V_l, V_r) consists of 4 edges $((1, 2), (1, 5), (3, 2), (3, 4))$, the maximum number of edge-disjoint alternating cycles in a cycle decomposition of $G(\gamma_n)$ is at most $\frac{4}{2} = 2$.

From Theorem 2, $d(\gamma_n) \geq b(\gamma_n) - c(\gamma_n) \geq n + 1 - 2 = n - 1$. On the other hand, $d(\gamma_n) \leq n - 1$ [WEHM82]. Finally, note that $d(\gamma_n^{-1}) = d(\gamma_n)$. $\square$

Before we prove that γ_n and γ_n^{-1} are the only permutations in S_n with a reversal distance of $n - 1$ (*strong Gollan conjecture*), we need to extend the concept of sorting permutations by reversals as follows: For any permutation of $\{1, \dots, n\}$, $\pi = \pi_1 \pi_2 \dots \pi_n$, let $\hat{\pi} = \hat{\pi}_1 \hat{\pi}_2 \dots \hat{\pi}_n$, with $\hat{\pi}_i = \pi_i + 1$, be a permutation of $\{2, 3, \dots, n+1\}$. Define $d(\hat{\pi})$ as the minimum number of reversals required to transform $\hat{\pi}$ to $234 \dots n+1$. Clearly, $d(\hat{\pi}) = d(\pi)$.

THEOREM 4 (strong Gollan conjecture). *For every n , γ_n and γ_n^{-1} are the only permutations that require $n - 1$ reversals to be sorted.*

Proof. Define $\mathcal{P}_n \equiv \{\pi | \pi \in S_n \text{ and } d(\pi) = n - 1\}$. We have seen that $\mathcal{P}_n \supseteq \{\gamma_n, \gamma_n^{-1}\}$. In what follows, we inductively prove that $\mathcal{P}_n \subseteq \{\gamma_n, \gamma_n^{-1}\}$.

For $n \leq 2$, the claim is trivial. Assume that the claim is true up to $n - 1$. Consider $\pi \in \mathcal{P}_n$. Let ρ_π be the unique reversal that brings n to the right end, that is, $\pi \cdot \rho_\pi = \pi' n$, where π' is a permutation of $\{1, \dots, n - 1\}$.

It follows that $d(\pi) \leq 1 + d(\pi')$. Then, $d(\pi') \geq d(\pi) - 1 = n - 2$. By induction, π' is either γ_{n-1} or γ_{n-1}^{-1} . Define $\mathcal{A} = \{\pi | \pi \cdot \rho_\pi = \gamma_{n-1} n\}$ and $\mathcal{B} = \{\pi | \pi \cdot \rho_\pi = \gamma_{n-1}^{-1} n\}$, where γn denotes the concatenation of a permutation γ with element n . Obviously, $\mathcal{P}_n \subseteq \mathcal{A} \cup \mathcal{B}$.

Likewise, define ρ'_π as the unique reversal that brings 1 to the left end, that is, $\pi \rho'_\pi = 1\pi'$, where π' is a permutation of $\{2, 3, \dots, n\}$. As before, $d(\pi') \geq n - 2$, which implies that π is $\hat{\gamma}_{n-1}$ or $\hat{\gamma}_{n-1}^{-1}$. Define $\mathcal{C} = \{\pi | \pi \cdot \rho'_\pi = 1\hat{\gamma}_{n-1}\}$ and $\mathcal{D} = \{\pi | \pi \cdot \rho'_\pi = 1\hat{\gamma}_{n-1}^{-1}\}$. Then, $\mathcal{P}_n \subseteq \mathcal{C} \cup \mathcal{D}$.

Therefore, $\mathcal{P}_n \subseteq \{\mathcal{A} \cup \mathcal{B}\} \cap \{\mathcal{C} \cup \mathcal{D}\}$. We state the following without proof:

1. $\mathcal{A} \cap \mathcal{C} = \phi$,
2. $\mathcal{A} \cap \mathcal{D} = \{\gamma_n\}$,
3. $\mathcal{B} \cap \mathcal{C} = \{\gamma_n^{-1}\}$,
4. $\mathcal{B} \cap \mathcal{D} = \phi$.

It follows that $\mathcal{P}_n \subseteq \{\gamma_n, \gamma_n^{-1}\}$. $\square$

4. Expected reversal distance. For any permutation $\pi \in S_n$, consider a set of cycles that form a maximum decomposition and partition them by size. Let $c_i(\pi)$ denote the number of alternating cycles of length i in a maximum decomposition, which do not include either vertex 0 or $n + 1$. There are at most two edge disjoint cycles, which contain the vertex 0 or $n + 1$. Then

$$(1) \quad c(\pi) \leq \sum_{i=4}^{2(n+1)} c_i(\pi) + 2.$$

For $k \leq 2(n + 1)$, let us consider cycles in the decomposition whose size is at least k . The number of such cycles is $c(\pi) - \sum_{i=4}^{k-1} c_i(\pi) - 2$. Now, the breakpoint graph of π has exactly $2b(\pi)$ edges. From this and the fact that the cycles are edge disjoint, we have

$$(2) \quad \forall k \leq 2(n + 1), \quad c(\pi) - \sum_{i=4}^{k-1} c_i(\pi) - 2 \leq \frac{1}{k} \left(2b(\pi) - \sum_{i=4}^{k-1} i c_i(\pi) \right).$$

From (1) and (2), we get

$$(3) \quad \forall k \leq 2(n + 1), \quad c(\pi) \leq \frac{1}{k} \left(2b(\pi) + \sum_{i=4}^{k-1} (k - i) c_i(\pi) \right) + 2.$$

Theorem 2 and inequality (3) imply that for all $k \leq 2(n + 1)$, we can bound $d(\pi)$ as

$$(4) \quad d(\pi) \geq \left(1 - \frac{2}{k} \right) b(\pi) - \frac{1}{k} \left(\sum_{i=4}^{k-1} (k - i) c_i(\pi) \right) - 2$$

$$(5) \quad \geq \left(1 - \frac{2}{k} \right) b(\pi) - \left(\sum_{i=4}^{k-1} c_i(\pi) \right) - 2.$$

Consider a permutation π chosen uniformly at random. As a cycle imposes a restriction on the permutation, it is intuitively clear that π does not have too many cycles. Denote the expected number of cycles of length i in a maximum cycle decomposition of $G(\pi)$ by $E(c_i(\pi)) = \frac{1}{n!} \sum_{\pi \in S_n} c_i(\pi)$. If we can bound $E(c_i(\pi))$, we can use (5) above to get a lower bound on the expected reversal distance. Lemma 2 provides such a bound which is, somewhat surprisingly, independent of n . Note that there is a slight ambiguity in the definition of $E(c_i(\pi))$, which depends on the choice of a maximum cycle decomposition for each $\pi \in S_n$. This does not affect Lemma 2, however, which holds for an arbitrary cycle decomposition.

LEMMA 2. $E(c_i(\pi)) \leq \frac{2^i}{i}$.

Proof. A cycle of length $i = 2t$ is a set of t breakpoints (unordered pairs of vertices) of the form

$$\{(x'_t, x_1), (x'_1, x_2), (x'_2, x_3), \dots, (x'_{t-1}, x_t)\}, \text{ with } x_j \sim x'_j.$$

Consider the set $x_1, x_2, \dots, x_t$. First, we claim that in every maximum cycle decomposition, $x_1, x_2, \dots, x_t$ are all distinct. To see this, consider the case $x_k = x_l$, for some $1 \leq k < l \leq t$. Then $(x'_k, x_{k+1}), (x'_{k+1}, x_{k+2}), \dots, (x'_{l-1}, x_l = x_k)$ form an alternating cycle, which can be detached to give a larger decomposition.

We have $\frac{n!}{(n-t)!}$ ways of selecting the ordered set, $x_1, x_2, \dots, x_t$. Once this is fixed we have a choice of at most 2 elements for each of the x'_j , giving a bound of $2^t \frac{n!}{(n-t)!}$ on the number of cycles of length $2t$. Note, however, that we count each $(2t)$ -cycle $2t$ times, so a tighter bound for the number of cycles of length $2t$ is $\frac{2^t}{2t} \frac{n!}{(n-t)!}$.

Choose an arbitrary $(2t)$ -cycle. The number of permutations in which this cycle can occur is no more than the number of ways of permuting the remaining $n - 2t$ elements plus the t pairs that form the cycle. Additionally, each pair can be flipped to give a different order, which gives at most $2^t(n - t)!$ permutations. Let p be the probability that an arbitrary $(2t)$ -cycle is present in a random permutation. Then $p \leq \frac{2^t(n-t)!}{n!}$ and

$$E(c_i(\pi)) = E(c_{2t}(\pi)) \leq \sum_{\{\text{all } (2t)\text{-cycles}\}} p \leq \frac{2^{2t}}{2t} = \frac{2^i}{i}. \quad \square$$

There are a total of $2n$ ordered adjacencies. Any such pair occurs in exactly $(n-1)!$ permutations, so the probability that it occurs in a random permutation is $\frac{1}{n}$. Then the expected number of adjacencies $E(a) = \frac{2n}{n} = 2$ and the expected number of breakpoints $E(b) = n + 1 - E(a) = n - 1$.

We use Lemma 2 and $E(b) = n - 1$ to get a bound on the expected diameter.

THEOREM 5. $E(d) \geq (1 - \frac{4.5}{\log n})n$.

Proof. From inequality (5), for all $k \leq 2(n + 1)$,

$$\begin{aligned} E(d) &\geq \left(1 - \frac{2}{k}\right) E(b) - \sum_{i=4}^{k-1} E(c_i) \geq \left(1 - \frac{2}{k}\right) (n - 1) - \sum_{i=4}^{k-1} \frac{2^i}{i} \\ &\geq \left(1 - \frac{2}{k}\right) (n - 1) - \sum_{i=4}^{k-1} 2^i \geq n - \frac{2n}{k} - \left(1 - \frac{2}{k}\right) - 2^k + 1 \\ &\geq n - \frac{2n}{k} - 2^k. \end{aligned}$$

TABLE 1
Comparison of theoretical and experimental lower bounds on expected diameter.

n	20	30	40	50	60	70	80	90	100
Theoretical	8.00	15.08	22.58	30.08	37.58	45.08	52.58	60.08	67.58
Experimental (matching)	12.60	19.92	27.34	34.05	41.09	48.73	55.80	63.33	70.49
Experimental (linear program)	12.6	20.8	28.5	35.9	43.6	51.7	58.9	67.6	74.2

Choose $k = \log \frac{n}{\log n}$. Then $2^k \leq \frac{n}{k}$ and

$$E(d) \geq \left(1 - \frac{3}{\log \frac{n}{\log n}}\right)n \geq \left(1 - \frac{4.5}{\log n}\right)n \quad \text{for } n \geq 2^{16}.$$

Lemma 2 and inequality (4) for $k = 10$ imply that $E(d) \geq (1 - \frac{4.5}{\log n})n$ for $19 < n < 2^{16}$. For $1 \leq n \leq 19$, $(1 - \frac{4.5}{\log n})n < 1$. $\square$

Although the bound provided by Theorem 5 is good asymptotically, it is weak for small values of n . However, the bound given by inequality (3) is tight if we select a k that gives the minimum value. We first give an example for $n = 100$ and then compare theoretical and experimental values for lower bounds on the expected diameter.

From Lemma 2 we have $E(c_4) \leq \frac{16}{4}$, $E(c_6) \leq \frac{64}{6}$, and $E(c_8) \leq \frac{128}{8}$. Consider a random permutation with $n = 100$ and $E(b) = n - 1 = 99$. Applying inequality (3) for $k = 8$, we derive $E(c) \leq \frac{1}{8} (198 + (8 - 4)4 + (8 - 6)\frac{64}{6}) + 2 \simeq 31.42$. This gives a lower bound for expected diameter with $n = 100$ as 67.58, which is close to experimental bounds based on maximum matching and linear programming [KS93]. Table 1 compares theoretical and experimental bounds over a range of n .

Kececioğlu and Sankoff [KS93] use the lower bounds to prune the branch and bound tree, in order to solve the reversal distance problem. In the case of signed reversals (see below), the cycle decomposition is unique, and we can use Theorem 2 to get comparable bounds, at a significant reduction in running time.

5. Short cycles and approximation of reversal distance. Starting from this section we discuss approximation algorithms for sorting by reversals. Define a *strip* of π as an interval $[i, j]$ such that $(i - 1, i)$ and $(j, j + 1)$ are breakpoints and no breakpoint lies between them. A strip is *increasing* if $\pi_i < \pi_j$; otherwise it is *decreasing*. A strip of one element is both increasing and decreasing, except for π_0 and π_{n+1} , which are always increasing.

A reversal can add or remove no more than two breakpoints. Define an i -reversal, $i \in \{0, 1, 2\}$, as one that removes i breakpoints. Our task is to remove breakpoints by reversals and merge strips into one increasing strip. Clearly, we need at least $\frac{b(\pi)}{2}$ reversals. For an upper bound on the number of reversals, Kececioğlu and Sankoff [KS93] give a greedy algorithm (Fig. 6) and prove the following assertions.

LEMMA 3. ([KS93]) *Let π be a permutation with a decreasing strip. Then (i) π has a 1- or 2-reversal, and (ii) if every reversal that removes a breakpoint of π leaves a permutation with no decreasing strips, then π has a 2-reversal.*

Partition a sequence of reversals into *rounds*, so that each round (except, perhaps, the first one) begins with a 0-reversal and has no other 0-reversals. Lemma 3 implies that in procedure *KS*, each round ends in a 2-reversal, thereby proving that every 0-reversal can be amortized against a 2-reversal and, *on the average*, we need at

Procedure $KS(\pi)$

while π contains a breakpoint **do**

$\rho = \text{Greedy}(\pi)$

$\pi = \pi \cdot \rho$

endwhile

Procedure $\text{Greedy}(\pi)$

begin

 Return a reversal that removes the most breakpoints of π ,
 resolving ties in favor of reversals that leave a decreasing strip.

end

FIG. 6. *The greedy algorithm.*

most one reversal to remove a breakpoint. Comparison of the upper bound of $b(\pi)$ reversals against the lower bound $\frac{b(\pi)}{2}$ provides a performance guarantee of 2. Can we do better?

Theorem 2 gives a stronger lower bound of $d(\pi) \geq b(\pi) - c(\pi)$, where $c(\pi)$ is the number of cycles in a maximum cycle decomposition of the breakpoint graph. Note that breakpoints correspond to black edges in this graph and every alternating cycle has at least 2 black edges. Therefore, the lower bound of $d(\pi) \geq \frac{b(\pi)}{2}$ is a simple corollary of Theorem 2.

Also, using inequality (2), we derive

$$d(\pi) \geq b(\pi) - c_4(\pi) - (c(\pi) - c_4(\pi)) \geq b(\pi) - c_4(\pi) - \frac{b(\pi) - 2c_4(\pi)}{3} = \frac{2}{3}b(\pi) - \frac{1}{3}c_4(\pi).$$

In the following sections we devise algorithms that sort π in at most $b(\pi) - \epsilon c_4(\pi)$ steps, for some $\epsilon > 0$. Then, the performance ratio of our algorithms is

$$(6) \quad \mathcal{A} = \max_{0 \leq c_4(\pi) \leq \frac{b(\pi)}{2}} \left\{ \frac{b(\pi) - \epsilon c_4(\pi)}{\frac{2}{3}b(\pi) - \frac{1}{3}c_4(\pi)} \right\} \leq \begin{cases} 2 - \epsilon, & \epsilon \leq \frac{1}{2}, \\ \frac{3}{2}, & \text{otherwise.} \end{cases}$$

6. Approximation algorithm for signed permutations. It is interesting to note that while the problem of sorting signed permutations is easier to handle, it is more relevant from a biological point of view. This is because genes are *directed* fragments of DNA sequences (Fig. 1).

We note that the concept of breakpoint graph as well as strips extends naturally to signed permutations. Define a transformation from a signed permutation π of order n to an unsigned permutation $\pi' \in S_{2n}$ as follows: replace $+i$ by $2i - 1, 2i$ and $-i$ by $2i, 2i - 1$. We observe that the identity signed permutation maps to the identity (unsigned) permutation, and the effect of a reversal on π can be mimicked by a reversal on π' . Therefore, any lower bound on π' is a lower bound on π . In particular, Theorem 2 holds.

For the upper bound, we shall perform reversals only across breakpoints so that any reversal on the unsigned permutation can be mimicked by a reversal on the signed permutation. It follows that, for our purpose, the two permutations are equivalent and in the following discussion, whenever we refer to the breakpoint graph or strips of a signed permutation, it is implied that we refer to the breakpoint graph or strips, respectively, of the transformed unsigned permutation.

Observe that in a breakpoint graph of signed permutations, every vertex has degree at most 2. Therefore the cycle decomposition is unique, thus making the case

Procedure *SignedSort*(π)

```

1. while  $\pi$  contains a breakpoint do
2.   if  $\pi$  has no decreasing strips
3.     if any 4-cycle  $C$  remains in  $G(\pi)$ 
4.       Find a cycle  $C'$  which crosses  $C$ .
5.       Do a 0-reversal on  $C'$  so that the 4-cycle  $C$  is oriented.
6.       Do a 2-reversal on the 4-cycle  $C$ .
7.     else
8.       Do a 0-reversal on an arbitrary cycle.
9.   else
10.     $\rho = \text{Greedy}(\pi)$ 
11.     $\pi = \pi \cdot \rho$ 
12. endwhile

```

FIG. 7. Algorithm for sorting signed permutations.

of signed permutations easier to handle. Below, we devise an algorithm that sorts signed permutations in $b(\pi) - \frac{1}{2}c_4(\pi)$ steps, thereby achieving a performance ratio of $\frac{3}{2}$. Later, we will use signed permutations to improve the performance guarantee for (unsigned) sorting by reversals.

In order to be able to sort a signed permutation π in less than $b(\pi)$ steps, we need 2-reversals that *do not have to be amortized* against 0-reversals. In the breakpoint graph of a signed permutation, 2-reversals correspond to elimination of 4-cycles, while 1-reversals correspond to shortening of longer cycles. However, the breakpoints might be oriented in such a way that 1- and 2-reversals are infeasible.

We call $\rho(i, j)$ a *reversal on a cycle* if the breakpoints (π_{i-1}, π_i) and (π_j, π_{j+1}) belong to the *same* cycle. A cycle is *oriented* if there exists a 1- or 2-reversal on it. Two cycles are *crossing* if some of the breakpoints corresponding to their black edges are *interleaved* in the permutation. For example, for γ_6 , $\rho : 315264 \rightarrow 315624$ is a reversal on the cycle $C = 134652$ (Fig. 3), since the breakpoints $(5, 2)$ and $(6, 4)$ belong to C . C is oriented since ρ is a 1-reversal. Cycles C_1 and C_2 in Fig. 3 are crossing as their breakpoints $((3, 1)$ and $(6, 4)$ in C_1 , $(1, 5)$ and $(4, 7)$ in C_2) are interleaved in the permutation 315264.

Note that a reversal on a cycle can orient an unoriented crossing cycle. The following lemma shows that we can use reversals on a cycle to orient 4-cycles, for signed permutations.

LEMMA 4. *Any 4-cycle C that is not oriented has a crossing cycle C' . Also, there exists a reversal on C' which will orient C .*

Proof. A 4-cycle that is not oriented is of the form $\dots i, j \dots j', i' \dots$, with $i \sim i'$ and $j \sim j'$. Since (j, j') is a gray edge in the breakpoint graph, the set of elements strictly between j and j' in the permutation, $\mathcal{S}$, is nonempty. Consider the largest and the smallest (not necessarily distinct) elements in $\mathcal{S}$. For at least one of these two elements, say k , there exists $k' \sim k$ such that $k' \notin \mathcal{S}$, and $k' \neq j, j'$. Consequently, there exist breakpoints (k, l) and (k', m) which are interleaved with (i, j) and (j', i') . Also, since all cycles in signed permutations are vertex-disjoint, $k' \neq i, i'$. Then, the cycle which contains the black edges (k, l) and (k', m) and the gray edge (k, k') is a crossing cycle for the 4-cycle C , and a reversal along the edges (k, l) and (k', m) will orient C . $\square$

Lemma 4 motivates the algorithm *SignedSort* for sorting signed permutations (Fig. 7). It uses the procedure *Greedy*, which was described earlier (Fig. 6).

LEMMA 5. *After step (6) in SignedSort, some decreasing strips remain.*

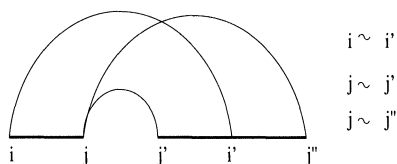


FIG. 8. A 2-overlapping pair of cycles (i, j, j', i') and (j, i, i', j'') .

Proof. Step (4) is executed when all strips in π are increasing. The 0-reversal of Step (5), which is on an increasing strip, creates a decreasing strip. To remove the decreasing strip, the reversal of Step 6 would need to be on the same interval, which it is not. $\square$

LEMMA 6. *If there is a 4-cycle in $G(\pi)$ at the beginning of any round of SignedSort (except, perhaps, the first one), then there are at least two 2-reversals in that round.*

Proof. If there is a 4-cycle in $G(\pi)$ at the beginning of any round (except, perhaps, the first one), that round begins with a 0-reversal followed by a 2-reversal. (Note that this cycle is nonoriented since all cycles in a permutation with no decreasing strips are nonoriented.) Also, from Lemma 5, some decreasing strips remain after this 2-reversal. At the point, as the permutation has decreasing strips we call *Greedy*. Then Lemma 3 implies that every round of *Greedy* ends in a 2-reversal, thus proving that as long as there are 4-cycles at the beginning of a round in *SignedSort*, there are at least two 2-reversals in that round. $\square$

THEOREM 6. *SignedSort sorts a signed permutation π in at most $b(\pi) - \frac{1}{2}c_4(\pi)$ reversals and provides an approximation ratio of $\frac{3}{2}$.*

Proof. Let d_i be the number of i -reversals in *SignedSort*. Observing that $b(\pi) = d_1 + 2d_2$, *SignedSort* sorts π in $d_0 + d_1 + d_2 = b(\pi) + d_0 - d_2$ steps.

Each round of *SignedSort* has exactly one 0-reversal (except, perhaps, the first one) and at least one 2-reversal, implying $d_2 \geq d_0$. In addition, by Lemma 6, if there is a 4-cycle in π at the beginning of a round, then there are at least two 2-reversals in that round. Therefore, if there are r such rounds, then $d_2 \geq d_0 + r$. On the other hand, each of the 4-cycles in $c_4(\pi)$ must be removed in a 2-reversal in one of these r rounds (4-cycles in signed permutations are unaffected by the other reversals). These $c_4(\pi)$ 2-reversals in r rounds together with 2-reversals in each of the remaining $d_0 - r$ rounds imply that $d_2 \geq d_0 + c_4(\pi) - r$.

From these two inequalities, we have $d_2 \geq d_0 + \frac{1}{2}c_4(\pi)$, which implies that *SignedSort* sorts π in $b(\pi) + d_0 - d_2 \leq b(\pi) - \frac{1}{2}c_4(\pi)$ steps. The bound on performance follows from equation (6). $\square$

7. Approximation algorithm for sorting by reversals. In general, finding a maximum cycle decomposition is not straightforward. In this section, we concentrate only on finding a cycle decomposition (not necessarily maximum) with a large number of 4-cycles, since such a decomposition will provide an improved performance ratio for sorting by reversals.

Any two 4-cycles can share at most two edges. Two 4-cycles are *2-overlapping* if they share two edges.

LEMMA 7. *If two 4-cycles of $G(\pi)$ are 2-overlapping, then one of them is oriented (i.e., a 2-reversal is possible on it).*

Proof. The only way a pair of alternating 4-cycles can 2-overlap is shown in Fig. 8. It can be verified that in any permutation of i, j, j', i', j'' that preserves such a structure, one of the 4-cycles is oriented. $\square$

The 4-cycle graph $H(\pi)$ of a permutation π is defined as a graph in which each

node corresponds to a 4-cycle of $G(\pi)$, and two nodes are connected by an edge if the corresponding 4-cycles share an edge in $G(\pi)$. An *independent set* in $H(\pi)$ corresponds to a set of edge disjoint 4-cycles. Furthermore, this graph has bounded degree, and there are efficient algorithms to find good approximations to a maximum independent set in bounded degree graphs.

We call a graph *strongly d -bounded* if the degree of every vertex in the graph is bounded by d and the degree of at least one vertex in every connected component of the graph is less than d .

LEMMA 8. *If $G(\pi)$ has no 2-overlapping cycles, then the 4-cycle graph $H(\pi)$ is strongly 4-bounded.*

Proof. It is easy to see that in a graph $G(\pi)$ without 2-overlapping cycles, a 4-cycle can have at most four other alternating cycles sharing edges with it. Consider a 4-cycle C containing the maximal element i of π among all 4-cycles in a given connected component of $H(\pi)$. Since $i + 1$ does not belong to any 4-cycle in this component, at most one of the neighboring cycles of C contains the vertex of $G(\pi)$ corresponding to that maximal element. It follows that C has at most three neighbors in $H(\pi)$. $\square$

LEMMA 9. *In a strongly d -bounded graph $G(V, E)$, an independent set of size at least $\frac{|V|}{d}$ can be computed in $O(E)$ time.*

Proof. Pick a vertex of degree less than d , add it to the independent set, and remove the neighbors. Clearly, the remaining graph is strongly d -bounded. Repeat this step until no vertices are left. As no more than d vertices are removed in each step, the size of the independent set is at least $\frac{|V|}{d}$. $\square$

LEMMA 10. *In a strongly d -bounded graph $G(V, E)$, a $\frac{2}{(d+1)}$ approximation to a maximum independent set can be computed in $O(E)$ time.*

Proof. Let the size of the maximum independent set be $a \cdot n$, for some a , where $|V| = n$. Consequently, the minimum vertex cover for G has a size $= (1 - a) \cdot n$. We can find a vertex cover V' of size at most $\min\{2(1 - a) \cdot n, n\}$ in $O(E)$ time [CLR90]. Then $I_1 = V \setminus V'$ is an independent set for G , of size at least $\max\{(2a - 1) \cdot n, 0\}$. Another independent set I_2 is given by Lemma 9 and is of size $\frac{n}{d}$. Obviously, we select the larger of the two sets.

There are two cases in analyzing the performance of our approximation:

$$\max\{(2a - 1)n, 0\} = 0:$$

This implies that $a \leq \frac{1}{2}$. In this case we select I_2 . Performance is $\frac{1}{ad} \geq \frac{2}{d}$.

$$\max\{(2a - 1)n, 0\} = (2a - 1)n:$$

In this case we select the larger of the two sets, I_1 and I_2 . The performance is $\max\{\frac{1}{da}, \frac{(2a-1)}{a}\}$, with $a > \frac{1}{2}$. In the worst case, $\frac{1}{ad} = \frac{(2a-1)}{a}$, which implies that $a = \frac{(d+1)}{2d}$, and performance is $\frac{2}{(d+1)}$. $\square$

Lemmas 7, 8, and 10 motivate the algorithm *ReversalSort* (Fig. 9). Theorem 7 provides a performance guarantee for this algorithm.

THEOREM 7. *The algorithm *ReversalSort* achieves an approximation ratio of $\frac{9}{5}$.*

Proof. Note that every reversal on the signed permutation σ' in step (4) of *ReversalSort* can be simulated on the unsigned permutation σ . Therefore, an upper bound on the number of reversals in *SignedSort*(σ') is an upper bound on the number of reversals in step (4) of *ReversalSort*. If the number of 4-cycles found in the cycle decomposition in step (2) is $c'_4(\sigma)$, then from Theorem 6, we have the bound $d(\sigma) \leq b(\sigma) - \frac{1}{2}c'_4(\sigma)$ for the number of reversals in step (4) of *ReversalSort*. Let $\rho_1\rho_2 \dots \rho_x$ be 2-reversals chosen in step (1) of the algorithm, so that $\pi\rho_1\rho_2 \dots \rho_x = \sigma$. Clearly, $b(\pi) - b(\sigma) = 2x$. Therefore, *ReversalSort* requires no more than $x + b(\sigma) - \frac{1}{2}c'_4(\sigma) = b(\pi) - x - \frac{1}{2}c'_4(\sigma)$ reversals to sort π .

Algorithm *ReversalSort*(π)

1. Starting with the permutation π , perform 2-reversals on $G(\pi)$ until no 2-overlapping cycles remain (Lemma 7). Let σ denote the resulting permutation.
2. Use the Independent Set approximation in $H(\sigma)$ to find a set of nonoverlapping 4-cycles of size at least $\frac{2}{5}c_4(\sigma)$ in the breakpoint graph $G(\sigma)$ (Lemmas 8 and 10). Find an arbitrary cycle decomposition of the remaining edges.
3. Split vertices of degree 4 in $G(\sigma)$ according to the cycle decomposition found in step (2), so that the cycles are vertex disjoint. In terms of strips, replace single elements by strips, oriented appropriately, resulting in a signed permutation σ' .
4. Call *SignedSort*(σ') to sort σ' . Sorting of σ' mimics sorting of σ .

FIG. 9. *Algorithm for sorting by reversals.*

Let $i(\delta)$ be the size of the maximum independent set in a 4-cycle graph $H(\delta)$. Lemmas 8 and 10 guarantee that $c'_4(\sigma) \geq \frac{2}{5}i(\sigma)$. On the other hand, $i(\pi) - i(\sigma) \leq 4x$, since every reversal in step (1) “destroys” at most four of the $i(\pi)$ vertices of the maximum independent set in $H(\pi)$ (there are at most 4 nonoverlapping cycles sharing edges with a 4-cycle). Therefore,

$$\begin{aligned} d(\pi) &\leq b(\pi) - x - \frac{1}{5}i(\sigma) &\leq b(\pi) - x - \frac{1}{5}(i(\pi) - 4x) \\ &\leq b(\pi) - \frac{1}{5}x - \frac{1}{5}c_4(\pi) &\leq b(\pi) - \frac{1}{5}c_4(\pi). \end{aligned}$$

The bound on performance follows from equation (6). $\square$

8. Improved approximation for sorting by reversals. In this section, we modify *ReversalSort* to improve the performance ratio. Recall that in step (1) of *ReversalSort*(π), we perform 2-reversals to transform π into a permutation σ , which has the property that $H(\sigma)$ is strongly 4-bounded. This allows us to find a set of 4-cycles of size at least $\frac{2}{5}c_4(\sigma)$. In this section, we transform π using 2-reversals into a permutation σ , so that $H(\sigma)$ is *bipartite*. Consequently, we can find a maximum set of nonoverlapping 4-cycles in σ , which leads to improved performance.

The problem of transforming π into ν is equivalent to sorting $\nu^{-1}\pi$, and, for convenience, we shall switch between the two notations. Denote $G(\pi, \nu) \equiv G(\nu^{-1}\pi)$, $b(\pi, \nu) \equiv b(\nu^{-1}\pi)$, $i(\pi, \nu) \equiv i(\nu^{-1}\pi)$. Observe that $G(\pi, \nu)$ and $G(\nu, \pi)$ coincide, but have reversed colors, i.e., black (gray) edges in $G(\pi, \nu)$ are gray (black) in $G(\nu, \pi)$. It follows that $b(\pi, \nu) = b(\nu, \pi)$ and $i(\pi, \nu) = i(\nu, \pi)$.

Figure 10 describes the procedure *Transform* that transforms π into a permutation σ with the following properties.

LEMMA 11. $H(\sigma)$ is *bipartite*.

Proof. Consider a 4-cycle in $G(\sigma)$ formed by the vertices $\sigma_i, \sigma_{i+1}, \sigma_j, \sigma_{j+1}$, with $i+1 < j$. Because no 2-reversals are possible on $\sigma = \text{Transform}(\pi)$, every 4-cycle in $G(\sigma)$ must be nonoriented. Therefore, we must have $\sigma_i \sim \sigma_{j+1}$ and $\sigma_j \sim \sigma_{i+1}$. Now, observe that these vertices form a 4-cycle in $G(\sigma^{-1})$ also, with the color on the edges reversed. If $\sigma_j - \sigma_{i+1} = \sigma_{j+1} - \sigma_i$, then the 4-cycle in $G(\sigma^{-1})$ is oriented. Therefore, $\sigma_j - \sigma_{i+1} = -(\sigma_{j+1} - \sigma_i)$.

To interpret this graphically, direct black edges of $G(\sigma)$ from π_k to π_{k+1} and gray edges from k to $k+1$. We call edges of a cycle *coordinated* if they are directed the same way along this cycle. Note that if black edges of a 4-cycle are not coordinated, then this cycle is oriented. Since there is no oriented cycles in $G(\sigma)$, black edges in every 4-cycle of $G(\sigma)$ are coordinated. Also, since there is no oriented cycles in $G(\sigma^{-1})$, black edges in every 4-cycle of $G(\sigma^{-1})$ are coordinated. An observation that

Procedure *Transform*(π)

begin

$\bar{\pi} = \pi$

$\bar{\nu} = \iota$ /* ι is the identity permutation */

while $\bar{\nu}^{-1} \cdot \bar{\pi}$ or $\bar{\pi}^{-1} \cdot \bar{\nu}$ has a 2-reversal, ρ

if ρ is a 2-reversal on $\bar{\nu}^{-1} \cdot \bar{\pi}$

$\bar{\pi} = \bar{\pi} \cdot \rho$

else $\bar{\nu} = \bar{\nu} \cdot \rho$

endwhile

return $\bar{\nu}^{-1} \cdot \bar{\pi}$

end

FIG. 10. Algorithm for preprocessing permutation π using 2-reversals.

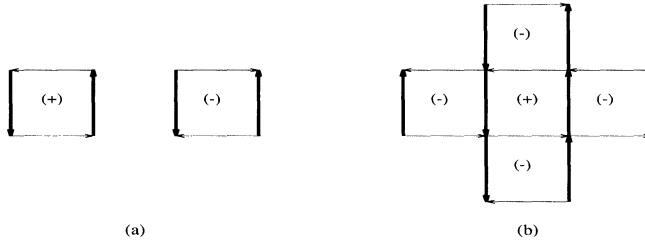


FIG. 11. (+) and (-) cycles in $G(\pi)$.

the edges of $G(\sigma)$ and $G(\sigma^{-1})$ have the same directions but reverse colors implies that gray edges in every 4-cycle of $G(\sigma)$ are coordinated. Only two such 4-cycles (denoted (+) and (-)) are possible (Fig. 11a). Further, observe that all cycles that share an edge with a cycle of type (+), must be of type (-), and vice-versa (Fig. 11b). This implies that $H(\sigma)$ is bipartite. $\square$

LEMMA 12. σ can be sorted in $b(\sigma) - \frac{1}{2}i(\sigma)$ reversals.

Proof. Because $H(\sigma)$ is a bipartite graph (Lemma 11), we can reduce maximum independent set problem in $H(\sigma)$ to the maximum matching problem [CLR90]. Moreover, since $H(\sigma)$ is a graph of bounded degree, we can find a maximum independent set in $H(\sigma)$ in $O(n^{\frac{3}{2}})$ time by the maximum matching algorithm [HK73]. This implies that we can find a cycle decomposition of $G(\sigma)$ in which the number of 4-cycles is $i(\sigma)$. The lemma then follows from Theorem 6. $\square$

Let $\rho_1 \rho_2 \dots \rho_x$ be the sequence of reversals in *Transform* that transforms π into $\bar{\pi}$ ($\bar{\pi} \rho_1 \rho_2 \dots \rho_x = \pi$) and $\varrho_1 \varrho_2 \dots \varrho_y$ be the sequence of reversals that transforms ι into $\bar{\nu}$ ($\bar{\nu} \varrho_1 \varrho_2 \dots \varrho_y = \iota$).

LEMMA 13. Let $\sigma = \text{Transform}(\pi)$. Then, $b(\pi) - b(\sigma) = 2(x + y)$, and $i(\pi) - i(\sigma) \leq 4(x + y)$.

Proof. In each iteration of the **while** loop in *Transform*, $(\bar{\pi}, \bar{\nu})$ is transformed into $(\bar{\pi}', \bar{\nu}')$, in one of two ways. Either (i) $\bar{\pi}' = \bar{\pi} \cdot \rho$, $\bar{\nu}' = \bar{\nu}$, or (ii) $\bar{\pi}' = \bar{\pi}$, $\bar{\nu}' = \bar{\nu} \cdot \rho$. In case (ii), ρ is a 2-reversal on $\bar{\pi}^{-1} \bar{\nu}$, which implies that $2 = b(\bar{\pi}^{-1} \bar{\nu}) - b((\bar{\pi}')^{-1} \bar{\nu}') = b(\bar{\nu}, \bar{\pi}) - b(\bar{\nu}', \bar{\pi}') = b(\bar{\pi}, \bar{\nu}) - b(\bar{\pi}', \bar{\nu}')$. A similar argument holds for case (i), implying that $b(\bar{\pi}, \bar{\nu}) - b(\bar{\pi}', \bar{\nu}') = 2$. $i(\bar{\pi}, \bar{\nu}) - i(\bar{\pi}', \bar{\nu}') \leq 4$ follows from the fact that every 2-reversal can destroy at most four nonoverlapping 4-cycles in the breakpoint graph.

The lemma follows from the fact that each reversal in *Transform* belongs to either the sequence $\rho_1 \rho_2 \dots \rho_x$ or the sequence $\varrho_1 \varrho_2 \dots \varrho_y$, implying a total of $x + y$ reversals. $\square$

ImprovedSort (Fig. 12) exploits the structure of the permutation $\sigma = \text{Trans-}$

Algorithm *ImprovedSort*(π)

1. Call *Transform*(π) to find a sequence of reversals $\rho_1\rho_2\ldots\rho_x$ and $\varrho_1\varrho_2\ldots\varrho_y$ such that $\pi\rho_1\rho_2\ldots\rho_x = \bar{\pi}$ and $\iota\varrho_1\varrho_2\ldots\varrho_y = \bar{\nu}$. Let $\sigma = \bar{\nu}^{-1}\bar{\pi}$.
2. Find a maximum set of nonoverlapping 4-cycles in $G(\sigma)$ by solving the maximum independent set problem in the bipartite graph $H(\sigma)$ (Lemma 11). Find an arbitrary cycle decomposition of the remaining edges.
3. Split vertices of degree 4 in $G(\sigma)$ according to the cycle decomposition found in step (2), so that the cycles are vertex disjoint. In terms of strips, replace single elements by strips, oriented appropriately, resulting in a signed permutation σ' .
4. Call *SignedSort*(σ') to find a sequence of reversals that sort σ' . Note that this sequence is mimicked by a sequence of reversals $\varphi_1\varphi_2\ldots\varphi_z$ that sorts σ .
5. Apply the sequence of reversals $\rho_1\rho_2\ldots\rho_x\varphi_1\varphi_2\ldots\varphi_z\varrho_y\ldots\varrho_2\varrho_1$ to sort π .

FIG. 12. *Improved algorithm for sorting by reversals.*

form(π) to sort π more efficiently. Theorem 8 analyzes the performance of this improved algorithm.

THEOREM 8. *ImprovedSort sorts π in $b(\pi) - \frac{1}{4}c_4(\pi)$ steps.*

Proof. From step (5) of *ImprovedSort*, $d(\pi) \leq x + y + z$. Lemmas 12 and 13 and the inequality $i(\pi) \geq c_4(\pi)$ provide an upper bound

$$\begin{aligned} x + y + z &\leq x + y + b(\sigma) - \frac{1}{2}i(\sigma) \\ &= b(\pi) - (x + y) - \frac{1}{2}i(\sigma) \\ &\leq b(\pi) - \frac{(i(\pi) - i(\sigma))}{4} - \frac{1}{2}i(\sigma) \\ &\leq b(\pi) - \frac{1}{4}i(\pi) \\ &\leq b(\pi) - \frac{1}{4}c_4(\pi). \quad \square \end{aligned}$$

Theorem 8 and equation (6) imply Corollary 1.

COROLLARY 1. *The algorithm ImprovedSort achieves an approximation ratio of $\frac{7}{4}$.*

9. Running time. We show that all our algorithms have complexity of $O(n^2)$. Consider lines 4–6 of *SignedSort*. If we maintain both π and π^{-1} , it takes $O(n)$ time to find a 4-cycle, as well as a crossing cycle. Kececioglu and Sankoff [KS93] give an $O(n)$ implementation of *Greedy*. Finally, no more than $n - 1$ reversals are required to sort a permutation π , which gives an upper bound of $O(n^2)$ for *SignedSort*. For *ReversalSort* and *ImprovedSort*, the most expensive preprocessing step is that of finding the maximum independent set in a bipartite graph, which takes $O(n^{\frac{3}{2}})$ time. Therefore, the time bounds do not change for the improved sorting algorithms.

10. Genome rearrangements. Sequence alignment is often the first step in molecular evolution studies. However, in many cases sequence alignment is very unreliable, thus making further evolutionary tree reconstruction almost impossible. For example, the similarity between many genes in herpes viruses is so low that it is frequently indistinguishable from the background noise. As a result, the classical methods of *sequence comparison* frequently lead to ambiguous conclusions for such highly diverged genomes, and alternative methods are sought (Karlin et al. [KMS94]). Since it is often found that the order of genes is much more conserved than the DNA sequence, an approach based on comparison of *gene orders* (Sankoff [S93]) versus traditional comparison of *DNA sequences* seems to be a method of choice for many “hard-to-analyze” genomes.

Surprisingly enough, genome comparison might also outperform gene comparison for such highly conserved genomes as plant mitochondrial DNA(mtDNA). The point mutation rate in plant mtDNA is estimated to be 100 times slower than in animal mtDNA, and many genes are 99%–99.9% identical in related species (Palmer and Herbon [PH88]). However, although there is little change in the DNA sequence of the plant mitochondrial genes from one species to another, there is rapid and extensive change in gene arrangements. This implies that the traditional methods for comparing sequences are not very conclusive in this case either.

The described algorithms for sorting by reversals were implemented and applied for analysis of genome rearrangements in herpes viruses, plant organelles, and mammalian X chromosomes. For plant organelles, Bafna and Pevzner [BP95] corrected previously postulated scenarios for genome rearrangements in the *Brassica* family. For herpes viruses, Hannenhalli et al. [HCKP95] were able to come out with three alternative gene orders in common herpesvirus ancestor. Surprisingly enough, in all biological examples we analyzed, $d(\pi) = b(\pi) - c(\pi)$, thus indicating that the bound provided by Theorem 2 might be rather tight. A similar observation has been made by Kececioğlu and Sankoff [KS94]: $b(\pi) - c(\pi)$ approximated $d(\pi)$ with accuracy 1 for each of 100 randomly chosen permutations on 10,000 elements. It indicates that the bound provided by the Theorem 2 is extremely tight and raises the problem of finding an exact estimate for reversal distance in terms of cycle decompositions.

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